

Testing Memory Under Cognitive Load Using the Mnemonic Similarity Task (MST)

Report 2: Full Inferential Analysis (H1–H9)

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Team Baby Bayes

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Abstract

This report presents the complete inferential analysis (H1–H9) for an MST study on how cognitive load at event boundaries affects recognition memory (REC) and lure discrimination (LDI). Report 2 revisits H1–H4 with corrected hypotheses and upgraded tests, and introduces five new analyses. All groups exceed chance on REC and LDI (H1–H2 supported). Boundary position does not affect LDI (H3 not supported) but does affect REC with a condition interaction (H7 partially supported). A lure-similarity gradient confirms pattern separation (H4 supported). Condition modulates LDI but not REC (H6 partially supported). A large boundary slowdown in encoding RT is found ($\eta_p^2=0.335$, H8 supported), with no speed–accuracy trade-off (H9 not supported).

1. Introduction

Event Segmentation Theory (EST) proposes that the brain briefly updates its internal “event model” at event boundaries, temporarily consuming cognitive resources and impairing encoding of immediately post-boundary items while strengthening memory for pre-boundary ones [1]. The hippocampus—particularly the dentate gyrus and CA3—supports pattern separation: storing similar experiences as distinct memory traces [2]. The Mnemonic Similarity Task (MST) probes this capacity, requiring participants to discriminate studied images from perceptually similar lures. This report covers H1–H9: H1–H4 are revisited from Report 1 with a structured format and upgraded analyses; H5–H9 are new.

2. Methods

2.1 Participants, Design, and Procedure

160 participants were split into three groups: `both_item_task` ($n=51$), `item_only` ($n=56$), and `task_only` ($n=53$). Boundaries were defined by: **item-only**—image-type switch (objects vs. scenes); **task-only**—question switch (e.g., “Indoors?” vs. “Shoebbox?”); **both**—simultaneous switches (highest load). Within 7-item blocks: Position 1 = Post-boundary, Positions 2–6 = Mid, Position 7 = Pre-boundary.

At test, participants classified 150 probes (60 Targets, 60 Lures in 5 similarity bins where Bin 1 = most similar, 30 Foils) as *Old*, *Similar*, or *New*.

2.2 Data Preprocessing

Raw data were cleaned by matching encoding/test CSVs on participant IDs, removing invalid rows, assigning boundary positions via `trial_index % 7` (0→Post, 6→Pre, else Mid), mapping lure filenames to inherit their target’s boundary label, and concatenating all participants into `compiled_preprocessed.csv`.

2.3 Memory Scores and Statistical Tests

$$\text{REC} = P(\text{Old} \mid \text{Target}) - P(\text{Old} \mid \text{Foil}), \quad \text{LDI} = P(\text{Similar} \mid \text{Lure}) - P(\text{Similar} \mid \text{Foil}).$$

Scores above zero indicate above-chance performance. All tests used Python (`scipy.stats`, `statsmodels`, `pingouin`), $\alpha = 0.05$; Holm’s correction used throughout.

3. Summary of Report 1 (H1–H4)

Report 1 conducted a preliminary inferential analysis covering H1–H4. Table 1 summarises the original tests and verdicts. Report 2 revisits all four with corrected hypotheses and upgraded tests; H3 receives an additional direction correction.

Table 1: Report 1 summary: original hypotheses, tests, key results, and verdicts.

H	Prediction	Test (Report 1)	Key Result	Verdict
H1	REC > 0 in all groups	One-sample one-tailed t -test	All $t > 24$, $p < 10^{-30}$	Supported
H2	LDI > 0 in all groups	One-sample one-tailed t -test	All $t > 11$, $p < 10^{-14}$	Supported
H3 [†]	LDI: Post > Mid > Pre	Three one-tailed paired t -tests	All $ \Delta < 0.011$, $p > 0.20$	Not Supported
H4	LDI increases Bin 1→5	Consecutive paired t -tests (Bonferroni + FDR)	2/4 steps significant	Partially Supported [‡]

[†]Direction revised in Report 2: Pre > Mid > Post per EST.

[‡]Upgraded to RM-ANOVA in Report 2; verdict revised to Supported.

Changes in Report 2: H3 direction corrected (Post > Mid > Pre → Pre > Mid > Post) and upgraded to Mixed ANOVA with Tukey HSD post-hoc; H4 upgraded from consecutive paired t -tests to one-way RM-ANOVA per condition (verdict upgraded from Partially Supported to Supported).

4. Results

H1: Recognition Memory Above Chance

Motivation. Before examining boundary or condition effects, we verify that each group’s mean REC score reliably exceeds zero.

Hypotheses.

$$\begin{aligned}
 H_0 : \quad & \mu_{\text{REC}} = 0 \quad (\text{performance at chance}) \\
 H_1 : \quad & \mu_{\text{REC}} > 0 \quad (\text{performance above chance})
 \end{aligned}$$

Tested separately per group.

Test and Justification. One-sample one-tailed t -test per group against zero. Population SD unknown; $n \geq 51$ (CLT applies); directional prediction justifies one-tailed test.

Results. All three groups recognised targets well above chance (Table 2). Mean REC: item-only = 0.634, both = 0.597, task-only = 0.596; all $p \leq 10^{-31}$.

Table 2: H1: One-sample one-tailed t -tests for REC > 0 by group.

Group	n	Mean REC	t	p-value
both_item_task	51	0.597	30.97	1.36×10^{-34}
item_only	56	0.634	30.96	8.67×10^{-37}
task_only	53	0.596	24.88	7.57×10^{-31}

Conclusion. H1 supported. All groups encode and recognise targets with high fidelity; astronomically small p -values rule out guessing. The slight item-only advantage is consistent with dual-task load diverting encoding resources, tested formally in H6.

H2: Pattern Separation Above Chance

Motivation. Above-chance LDI confirms that hippocampal pattern-separation mechanisms function under all three load conditions, establishing the validity of the core MST measure.

Hypotheses.

$$H_0 : \mu_{\text{LDI}} = 0 \quad (\text{lure discrimination at chance})$$

$$H_1 : \mu_{\text{LDI}} > 0 \quad (\text{lure discrimination above chance})$$

Tested separately per group.

Test and Justification. One-sample one-tailed t -test per group (same rationale as H1: unknown SD, $n \geq 51$, directional prediction).

Results. All groups discriminated lures above chance (Table 3). Mean LDI: task-only = 0.381, item-only = 0.319, both = 0.270.

Table 3: H2: One-sample one-tailed t -tests for LDI > 0 by group.

Group	n	Mean LDI	t	p-value
both_item_task	51	0.270	11.04	2.55×10^{-15}
item_only	56	0.319	14.09	3.38×10^{-20}
task_only	53	0.381	14.74	1.67×10^{-20}

Conclusion. H2 supported. The dual-task group’s markedly lower LDI ($\bar{x}$ =0.270 vs. 0.381 for task-only) suggests that simultaneous boundary changes reduce hippocampal encoding precision; the between-group difference is tested formally in H6.

H3: Does Boundary Position Affect LDI?

Revision from Report 1

Original (Report 1): Post > Mid > Pre on LDI, tested with three one-tailed paired t -tests, assuming boundaries *boost* post-boundary LDI. **Correction (Report 2):** EST predicts that context-updating at a boundary *impairs* post-boundary encoding; the correct directional prediction is Pre > Mid > Post. Test upgraded to a 3×3 Mixed ANOVA (position × condition) with Tukey HSD post-hoc.

Motivation. EST predicts a boundary-position gradient in mnemonic precision: pre-boundary items are encoded in a stable event context and should yield higher LDI; post-boundary items are encoded during context updating and should yield lower LDI.

Hypotheses.

$$H_0 : \mu_{\text{LDI,Pre}} = \mu_{\text{LDI,Mid}} = \mu_{\text{LDI,Post}} \quad (\text{no boundary-position effect on LDI})$$

$$H_1 : \mu_{\text{LDI,Pre}} > \mu_{\text{LDI,Mid}} > \mu_{\text{LDI,Post}} \quad (\text{pre-boundary advantage; corrected from Report 1})$$

Test and Justification. A 3×3 Mixed ANOVA (within: boundary position; between: condition; n =160) jointly estimates main effects and their interaction. Assumption checks: normality (Shapiro–Wilk p >0.05 in all cells); sphericity tenable (Mauchly W >0.96, p >0.34); homogeneity of variance confirmed (Levene p >0.16). Tukey HSD post-hoc tests followed up the significant condition main effect.

Results. The Mixed ANOVA found no significant main effect of boundary position ($F(2, 314)$ =0.28, p =0.753, η_p^2 =0.002) and no interaction ($F(4, 314)$ =1.45, p =0.218; Table 4). The condition main effect was significant ($F(2, 157)$ =5.50, p =0.005, η_p^2 =0.065); Tukey HSD identified only **both_item_task** vs. **task_only** as significantly different (Δ =−0.110, p =0.003, g =−0.635; Table 5).

Table 4: H3: Mixed ANOVA for LDI (boundary position × condition).

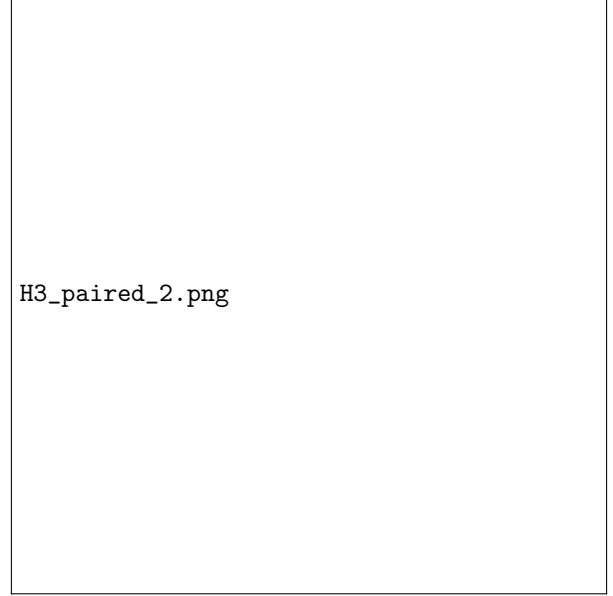
Source	DF1	DF2	F	p	η_p^2
Condition	2	157	5.498	0.005	0.065
Boundary Position	2	314	0.284	0.753	0.002
Position × Cond.	4	314	1.449	0.218	0.018

Table 5: H3: Tukey HSD post-hoc for the condition main effect on LDI.

Comparison	Mean Diff	p (Tukey)	Hedges' g	Sig.
both_item_task vs. item_only	−0.049	0.337	−0.283	No
both_item_task vs. task_only	−0.110	0.003	−0.635	Yes
item_only vs. task_only	−0.061	0.197	−0.359	No



(a) LDI distributions by boundary position.



(b) p -values for directional comparisons; none reaches $\alpha=0.05$.

Figure 1: H3: Boundary position does not reliably shift LDI across any analysis.

Conclusion. H3 not supported. The Mixed ANOVA finds no position effect on LDI; grand means (Pre = 0.323, Mid = 0.330, Post = 0.319) are effectively flat. Pattern separation appears driven by overall encoding quality (captured by condition) rather than fine-grained item position. High individual variability ($SD \approx 0.17\text{--}0.19$) and block-modulo boundary assignment likely attenuate any true positional signal.

H4: Do More Similar Lures Lead to Lower LDI?

Motivation. Lures span 5 similarity bins (Bin 1 = most similar to Bin 5 = least). If hippocampal pattern separation operates as theorised [2], rejection difficulty should decrease monotonically with bin number, validating the stimulus design.

Hypotheses.

$$\begin{aligned}
 H_0 : \quad & \mu_{\text{LDI}, \text{Bin1}} = \mu_{\text{LDI}, \text{Bin2}} = \dots = \mu_{\text{LDI}, \text{Bin5}} \quad (\text{no similarity gradient}) \\
 H_1 : \quad & \mu_{\text{LDI}, \text{Bin1}} < \mu_{\text{LDI}, \text{Bin2}} < \dots < \mu_{\text{LDI}, \text{Bin5}} \quad (\text{monotonic similarity gradient})
 \end{aligned}$$

Test and Justification. One-way RM-ANOVA per condition (within-subjects factor: bin, 5 levels) with Bonferroni-corrected pairwise post-hoc tests. RM-ANOVA is appropriate because each participant provides LDI at all five bins, accounting for individual-level variation. Sphericity held for all conditions (Table 6); no Greenhouse–Geisser correction required.

Table 6: H4: Mauchly’s sphericity test ($\text{LDI} \sim \text{bin}$, per condition).

Condition	Mauchly’s W	χ^2	p	Sphericity?
both_item_task	0.784	11.77	0.227	Yes
item_only	0.886	6.46	0.693	Yes
task_only	0.772	13.06	0.160	Yes

Results. All three conditions showed a highly significant effect of bin on LDI (Table 7; Figure 4), with medium effect sizes ($\eta_g^2 = 0.062\text{--}0.109$).

Table 7: H4: One-way RM-ANOVA results per condition ($\text{LDI} \sim \text{lure similarity bin}$).

Condition	F	df _{num}	df _{den}	p	η_g^2
both_item_task	10.098	4	200	< 0.001	0.062
item_only	17.349	4	220	< 0.001	0.096
task_only	20.639	4	208	< 0.001	0.109

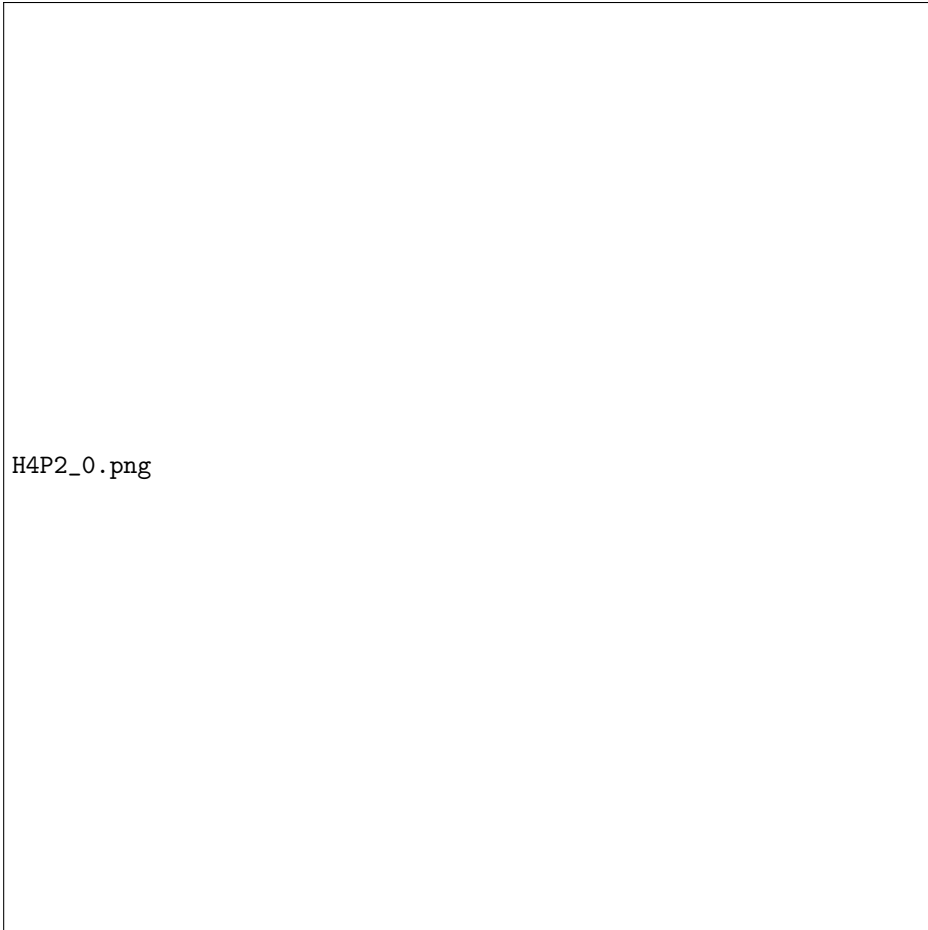


Figure 2: H4: Mean LDI per bin by condition ($\pm \text{SEM}$). Bin 1 is consistently lowest; task-only shows the richest gradient.

Post-hoc Bonferroni tests ($p_{\text{bonf}} < 0.05$; Table 8) show Bin 1 as the dominant outlier across all conditions. Mean LDI per condition $\times$ bin: **both**: 0.175, 0.278, 0.249, 0.319, 0.320; **item**: 0.193, 0.342, 0.335, 0.369, 0.382; **task**: 0.249, 0.344, 0.396, 0.445, 0.471.

Table 8: H4: Significant Bonferroni post-hoc bin comparisons per condition ($p_{\text{bonf}} < 0.05$).

Condition	Significant pairs	Pattern
both_item_task	Bin1<Bin2 ($p=0.021$), Bin1<Bin4 ($p<0.001$), Bin1<Bin5 ($p<0.001$), Bin3<Bin5 ($p=0.015$)	Bin 1 outlier; Bins 2–5 plateau
item_only	Bin1<Bin2, Bin1<Bin3, Bin1<Bin4, Bin1<Bin5 (all $p<0.001$)	Sharp step from Bin 1; flat after
task_only	Bin1<Bin2 ($p=0.006$), Bin1<Bin3,4,5 ($p<0.001$), Bin2<Bin4 ($p=0.012$), Bin2<Bin5 ($p=0.004$), Bin3<Bin5 ($p=0.035$)	Near-monotonic gradient across all 5 bins

Conclusion. H4 supported. LDI increases significantly from Bin 1 toward higher bins in all conditions, confirming that perceptual similarity drives pattern-separation difficulty. The gradient is not uniformly monotonic: Bin 1 is a clear outlier in `both_item_task` and `item_only` with Bins 2–5 largely plateauing; only `task_only` approaches true monotonicity. Once lures are moderately distinct, pattern separation approaches ceiling, with the ceiling reached sooner under higher cognitive load.

H5: Does Test-Phase Response Time Vary with Boundary Position?

Motivation. If boundary processing produces weaker post-boundary memory traces, participants may respond more slowly at test when probed with those items, providing a retrieval-stage signature of encoding disruption.

Hypotheses.

$$\begin{aligned}
 H_0 : \quad & \mu_{\text{RT}_{\text{test}},\text{Post}} \leq \mu_{\text{RT}_{\text{test}},\text{Mid}} \quad \text{and} \quad \mu_{\text{RT}_{\text{test}},\text{Post}} \leq \mu_{\text{RT}_{\text{test}},\text{Pre}} \\
 H_1 : \quad & \mu_{\text{RT}_{\text{test}},\text{Post}} > \mu_{\text{RT}_{\text{test}},\text{Mid}} \quad \text{and} \quad \mu_{\text{RT}_{\text{test}},\text{Post}} > \mu_{\text{RT}_{\text{test}},\text{Pre}}
 \end{aligned}$$

Test and Justification. Two one-tailed paired t -tests (Post > Mid; Post > Pre) on mean test-phase RT per participant ($n=158$). Paired to control for individual RT differences; one-tailed given the directional prediction.

Results. Neither comparison reached significance (Table 9). Post-boundary RT was marginally *shorter* than Mid ($\Delta=-0.037$ s, $p=0.816$); grand means: Pre = 2.37 s, Mid = 2.44 s, Post = 2.41 s.

Table 9: H5: One-tailed paired t -tests for boundary RT differences (test phase).

Comparison	Mean Diff (s)	t	p-value	Supported
Post > Mid	-0.037	-0.903	0.816	No
Post > Pre	+0.034	+0.933	0.176	No

Conclusion. H5 not supported. Test-phase RT is not shaped by boundary position, suggesting any encoding-time cost is transient and does not persist as a lasting feature of the memory trace accessible at retrieval.

H6: Do Conditions Differ in Overall REC and LDI?

Motivation. If cognitive load selectively impairs hippocampal encoding precision, LDI should differ across conditions while coarser recognition (REC) remains stable—a dissociation informative about the neural mechanisms involved.

Hypotheses. For each metric $M \in \{\text{REC}, \text{LDI}\}$:

$$\begin{aligned}
 H_0 : \quad & \mu_{M,\text{both}} = \mu_{M,\text{item}} = \mu_{M,\text{task}} \quad (\text{no condition effect on } M) \\
 H_1 : \quad & \text{at least one pair of conditions differs significantly on } M
 \end{aligned}$$

Test and Justification. Three pairwise Welch t -tests per metric (Holm-corrected within each metric). Welch’s test is preferred over one-way ANOVA as it does not assume equal variances across conditions.

Results. REC: no pairwise comparison survived Holm correction (all $p_{\text{Holm}} > 0.26$). LDI: `both_item_task` vs. `task_only` significant ($\Delta = -0.110$, $t = -3.045$, $p_{\text{Holm}} = 0.009$; Figure 6; Table 10).

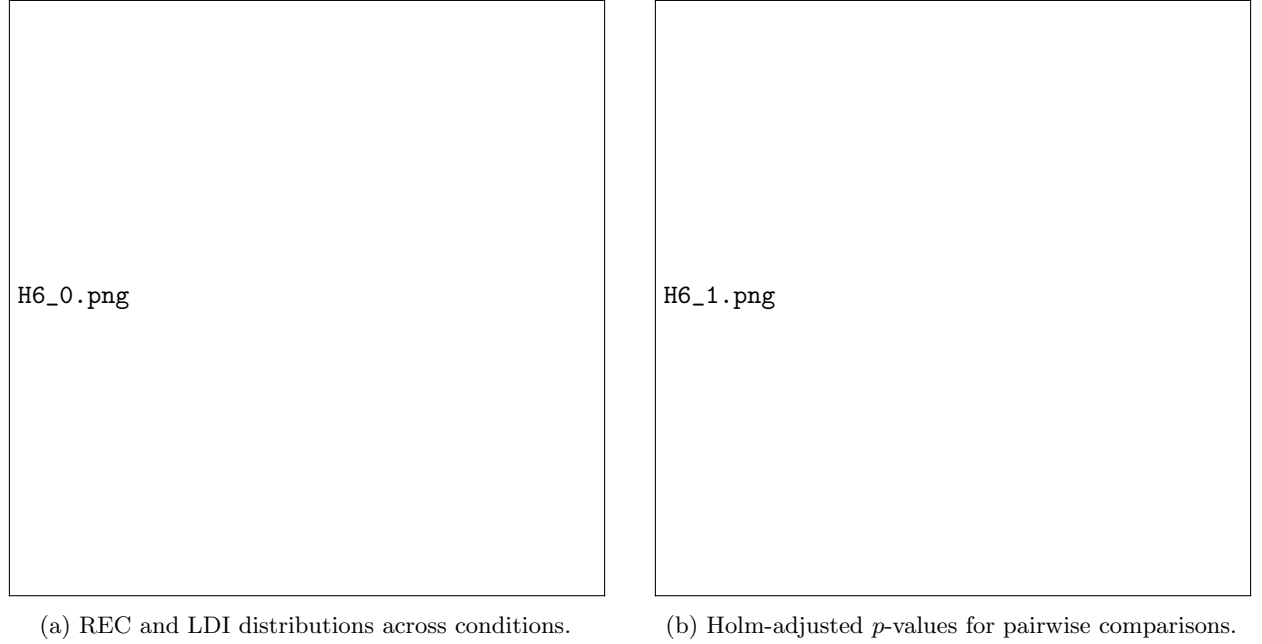


Figure 3: H6: Condition significantly affects LDI (dual-task < task-only) but not REC.

Table 10: H6: Pairwise Welch t -tests for condition differences (Holm-corrected).

Metric	Comparison	Mean Diff	t	p_{raw}	p_{Holm}	Sig.
REC	both vs. task	+0.006	+0.197	0.844	0.844	No
	both vs. item	-0.032	-1.121	0.265	0.691	No
	task vs. item	-0.038	-1.207	0.230	0.691	No
LDI	both vs. task	-0.110	-3.045	0.003	0.009	Yes
	both vs. item	-0.049	-1.432	0.155	0.155	No
	task vs. item	+0.061	+1.789	0.076	0.153	No

Conclusion. **H6 partially supported** (LDI affected; REC not). Task-only showed reliably higher mnemonic discrimination than dual-task ($p_{\text{Holm}} = 0.009$), confirming that simultaneous item-type and task-switch demands selectively degrade the fine-grained hippocampal encoding needed for pattern separation while leaving coarser recognition intact.

H7: Does Boundary Position Affect REC, and Does Condition Moderate This?

Motivation. H3 found no boundary-position effect on LDI. Recognition (REC) operates via different mechanisms and may reveal a boundary effect, potentially amplified by condition load.

Hypotheses.

- H_0^{pos} : boundary position has no main effect on REC
- H_0^{int} : the REC pattern across positions is identical across conditions
- H_1 : position significantly affects REC and/or is modulated by condition

Test and Justification. Mixed-Model ANOVA on per-participant REC by boundary position (within: position; between: condition; $n=160$). Sphericity tenable (Mauchly $W=0.973$, $\chi^2(2)=4.37$, $p=0.113$); Levene $p > 0.57$ for all cells.

Results. Position main effect significant ($F(2, 314)=4.64$, $p=0.010$, $\eta_p^2=0.029$); condition not significant ($F(2, 157)=0.976$, $p=0.379$); interaction significant ($F(4, 314)=2.99$, $p=0.019$, $\eta_p^2=0.037$; Table 11; Figure 7). Post-hoc Bonferroni: Mid > Post overall ($t(159)=2.641$, $p_{\text{bonf}}=0.027$); within **both_item_task**: Mid > Post ($p_{\text{bonf}}=0.003$) and Post < Pre ($p_{\text{bonf}}=0.022$).

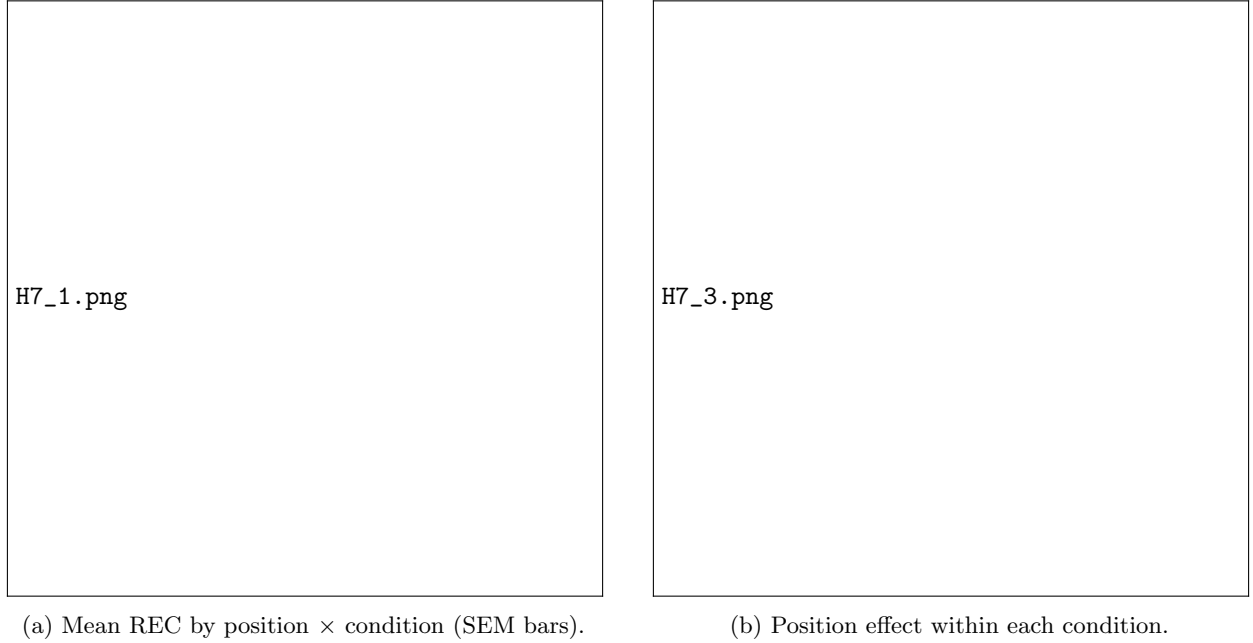


Figure 4: H7: Post-boundary REC drop is largest under dual-task load.

Table 11: H7: Mixed-Model ANOVA for REC (position $\times$ condition).

Source	DF1	DF2	F	p	η_p^2
Condition	2	157	0.976	0.379	0.012
Boundary Position	2	314	4.635	0.010	0.029
Position $\times$ Cond.	4	314	2.989	0.019	0.037

Conclusion. H7 partially supported. REC is reliably lower at post-boundary than mid-boundary positions, and the drop is amplified under maximal load (**both_item_task**), consistent with EST. The absence of an analogous LDI effect (H3) suggests recognition and pattern separation respond differently to boundary-related encoding disruption.

H8: Does Encoding RT Show a Boundary Slowdown?

Motivation. If boundaries trigger a mandatory context-update process [1], post-boundary items should elicit longer encoding RT, providing direct behavioural evidence for the EST mechanism independent of memory outcomes.

Hypotheses.

$$\begin{aligned}
 H_0 : \quad & \mu_{\text{RT}_{\text{enc}}, \text{Post}} = \mu_{\text{RT}_{\text{enc}}, \text{Pre}} \quad (\text{no boundary slowdown}) \\
 H_1 : \quad & \mu_{\text{RT}_{\text{enc}}, \text{Post}} > \mu_{\text{RT}_{\text{enc}}, \text{Pre}} \quad (\text{post-boundary encoding is slower})
 \end{aligned}$$

Test and Justification. Mixed-Model ANOVA on per-participant mean encoding RT (within: Pre vs. Post; between: condition; $n=160$), simultaneously estimating main boundary effect and condition-dependent modulation.

Results. All three ANOVA effects were highly significant (Figure 8; Table 12): boundary position $F(1, 157)=78.94$, $p<0.0001$, $\eta_p^2=0.335$ (large); condition $F(2, 157)=11.65$, $p<0.0001$, $\eta_p^2=0.129$; interaction $F(2, 157)=38.22$, $p<0.0001$, $\eta_p^2=0.327$.

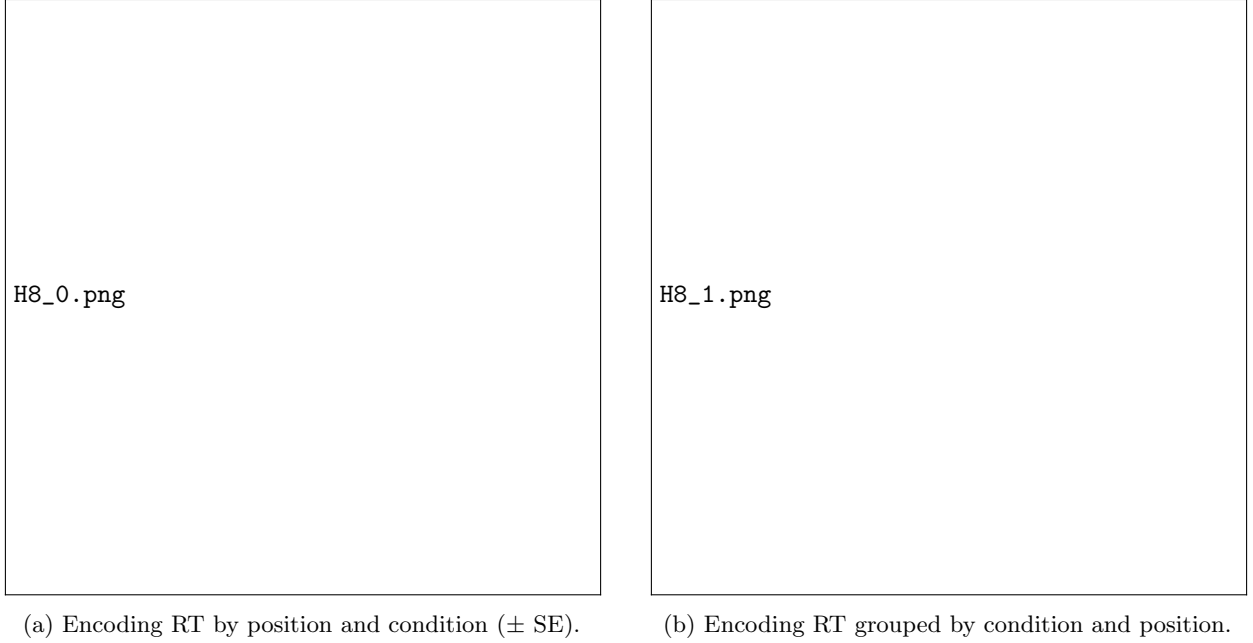


Figure 5: H8: Large boundary slowdown ($\eta_p^2=0.335$) with a significant condition interaction.

Table 12: H8: Mixed-Model ANOVA for encoding RT (Pre vs. Post).

Source	DF1	DF2	F	p	η_p^2
Condition	2	157	11.651	< 0.0001	0.129
Boundary Position	1	157	78.941	< 0.0001	0.335
Interaction	2	157	38.224	< 0.0001	0.327

Conclusion. H8 fully supported. The boundary slowdown ($\eta_p^2=0.335$) is the strongest effect in the dataset, directly validating EST. The large interaction ($\eta_p^2=0.327$) indicates that dual-task boundaries trigger the most resource-intensive context update. The absence of an RT–memory link (H9) suggests this slowdown is a mandatory, memory-neutral context-update obligation rather than an elaborative encoding strategy.

H9: Speed–Accuracy Trade-off (Encoding RT vs. REC & LDI)

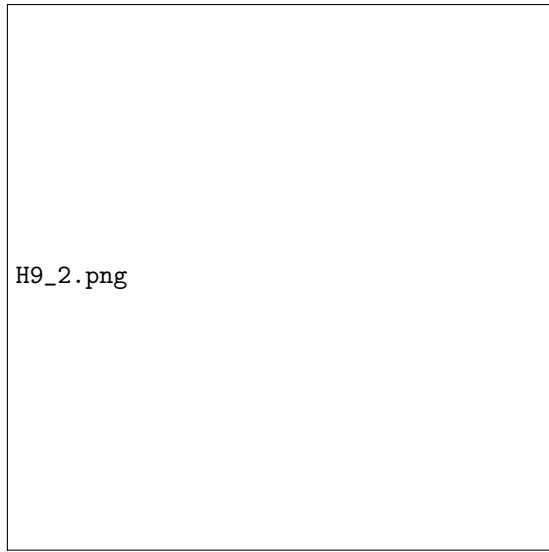
Motivation. The large encoding RT slowdown at boundaries (H8) raises a natural question: does slower encoding predict better or worse memory?

Hypotheses.

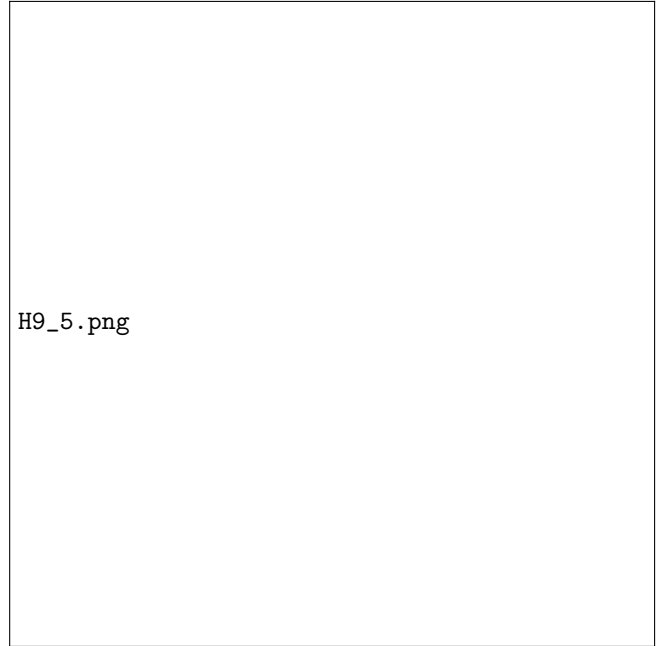
$$\begin{aligned}
H_0 : \quad & \rho(\text{RT}_{\text{enc}}, M) = 0 \quad \text{for } M \in \{\text{REC}, \text{LDI}\}, \text{ per condition} \\
H_1 : \quad & \rho(\text{RT}_{\text{enc}}, M) \neq 0 \quad (\text{a significant speed–accuracy relationship exists})
\end{aligned}$$

Test and Justification. Pearson r between mean encoding RT and memory scores per condition (six two-tailed tests). Spearman ρ computed for `task_only` as a robustness check.

Results. No correlation was significant in any condition (Figure 9; Table 13). In `both.item.task` the RT–REC correlation was weakly negative ($r=-0.255$, $p=0.071$). Spearman checks for `task_only`: $\rho=0.125$ ($p=0.372$, REC) and $\rho=0.171$ ($p=0.222$, LDI).



(a) Heatmap of Pearson r (encoding RT $\times$ REC/LDI).



(b) Correlation coefficients with 95% CI.

Figure 6: H9: No speed–accuracy trade-off; encoding RT does not predict memory quality.

Table 13: H9: Pearson correlations between encoding RT and memory scores by condition.

Condition	n	Metric	r	p-value	Significant
both_item_task	51	RT vs. REC	−0.255	0.071	No
		RT vs. LDI	−0.196	0.168	No
item_only	56	RT vs. REC	+0.033	0.807	No
		RT vs. LDI	−0.178	0.190	No
task_only	53	RT vs. REC	+0.072	0.608	No
		RT vs. LDI	+0.204	0.142	No

Conclusion. H9 not supported. Encoding speed does not predict memory quality within any condition. Individual RT differences likely reflect domain-general processing speed rather than encoding-strategy variation, consistent with the boundary slowdown (H8) being a mandatory context-update obligation that consumes time but does not alter memory trace quality.

5. Summary of All Hypotheses

Table 14: Complete summary of all nine hypotheses: predictions, tests, key statistics, and verdicts.

H	Prediction	Test (Report 2)	Key Statistics	Verdict
H1	REC > 0 across groups	One-sample one-tailed t	All $t > 24$, $p < 10^{-30}$	Supported
H2	LDI > 0 across groups	One-sample one-tailed t	All $t > 11$, $p < 10^{-14}$	Supported
H3 [†]	LDI: Pre > Mid > Post	Mixed ANOVA + Tukey HSD	Position: $F(2, 314)=0.28$, $p=0.753$	Not Supported
H4	Bin LDI increases Bin 1→5	RM-ANOVA per condition + Bonferroni	All $F > 10$, $p < 0.001$; $\eta_p^2=0.06-0.11$	Supported
H5	Test RT: Post > Pre/Mid	One-tailed paired t	$\Delta=-0.037$ s & $+0.034$ s, $p>0.17$	Not Supported
H6	Condition differences in REC & LDI	Welch t (Holm)	LDI: dual-task < task-only ($p_{\text{Holm}}=0.009$); REC: n.s.	Partially Supported
H7	Position $\times$ condition on REC	Mixed ANOVA	$F_{\text{pos}}=4.64$, $p=0.010$; $F_{\text{int}}=2.99$, $p=0.019$	Partially Supported
H8	Encoding RT: Post > Pre	Mixed ANOVA	$F(1, 157)=78.94$, $p<0.0001$, $\eta_p^2=0.335$	Supported
H9	Encoding RT correlates with memory	Pearson r per condition	All $ r < 0.26$, all $p > 0.07$	Not Supported

[†]Direction revised from Post > Mid > Pre (Report 1) to Pre > Mid > Post (Report 2).

6. Conclusion and Future Directions

Memory is robust but selectively load-sensitive (H1, H2, H6). All groups performed well above chance on both REC and LDI. The dual-task condition scored significantly lower on LDI than task-only ($p_{\text{Holm}}=0.009$, H6), confirming that simultaneous boundary changes selectively impair hippocampal pattern separation without affecting coarser recognition. This dissociation suggests pattern separation has a lower capacity ceiling under dual cognitive load.

Boundary position matters for REC but not LDI (H3, H7). H3 found no position effect on LDI across any analysis. H7’s Mixed ANOVA on REC revealed a significant position main effect ($p=0.010$) and condition interaction ($p=0.019$): mid-boundary REC exceeded post-boundary REC, driven primarily by `both_item_task`. This asymmetry suggests recognition and pattern separation respond differently to boundary encoding disruption.

Encoding RT robustly captures the boundary cost (H8, H5, H9). The strongest signal in the dataset is the boundary slowdown in encoding RT ($\eta_p^2=0.335$, $F=78.94$, $p<0.0001$), directly validating EST. The large interaction ($\eta_p^2=0.327$) confirms dual-task boundaries are most resource-intensive. Absent test-phase RT effects (H5) and absent RT–memory correlations (H9) together indicate the slowdown is a mandatory, memory-neutral context-update process.

Similarity gradient is real but non-uniform (H4). RM-ANOVA confirmed a significant bin effect in all conditions ($p<0.001$). Bin 1 is a consistent outlier while Bins 2–5 plateau in `both_item_task` and `item_only`; only `task_only` shows a richer near-monotonic gradient. Future work could use pupillometry or EEG to verify the block-modulo boundary assignment, and mediation analyses could test whether the encoding RT slowdown (H8) partially mediates the REC boundary effect (H7).

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Contributions

Name	Roll No.	Contributions
Achyutananda Sahoo	2025202028	EDA, H4 follow-up, H7 ANOVA, H8 analysis, H3 ANOVA revision
Satyajit Priyadarshi	2025201055	Data preprocessing, H1–H3, H5, H6, H9
Ameya Purohit	2025202006	Hypothesis testing, p -value correction, interpretation